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Multi-view feature consistency learning for cross-domain open-set mechanical fault diagnosis

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ABSTRACT

Deep learning-enabled cross-domain fault diagnosis plays a crucial role in improving the operational reliability and safety of mechanical systems. Conventional approaches usually assume that the testing data share an identical label space with the training data, implying consistent fault types across different engineering scenarios. However, in practical applications, fault types encountered during deployment are often unpredictable, and novel faults may arise, thereby undermining the effectiveness of existing domain adaptation methods. To address these challenges, a novel open-set domain adaptation framework termed multi-view feature consistency learning (MFCL) is proposed. Specifically, source-domain prototype constraints and domain-adversarial learning are jointly employed to align feature distributions between source and target domains, while multi-view contrastive learning enhances the representational quality of multi-view unlabeled target samples. Furthermore, an entropy-based adversarial separation mechanism is introduced to enable accurate identification of unknown classes during adaptation, thereby facilitating the separation between known and unknown classes. In addition, an unknown-aware consistency learning strategy is developed by integrating prototype-guided pseudo-labeling with multi-view features, which further boosts cross-domain classification performance. Extensive experiments on the SDUST and BJTU-RAO bearing datasets demonstrate that the proposed method outperforms state-of-the-art approaches in open-set domain adaptation and holds strong potential for enhancing the reliability of mechanical systems in practice.

1. Introduction

Accurate health assessment of rotating machinery is pivotal for maintaining the stability and reliability of industrial systems [1]. In recent years, data-driven diagnostic approaches have attracted considerable attention due to their ability to capture complex relationships between operating conditions and fault characteristics from large-scale monitoring data [2]. Within this context, deep learning has become a central research focus in fault diagnosis, owing to its strong feature extraction capability and end-to-end learning advantages [3–5]. Unlike traditional methods that rely heavily on expert knowledge and manual feature engineering, deep learning models automatically learn advanced feature representations directly from raw signals, substantially improving both diagnostic accuracy and efficiency [6,7]. Representative architectures, including CNNs [8,9], GNNs [10,11], and Transformers [12,13], have all demonstrated remarkable performance in various fault diagnosis tasks.

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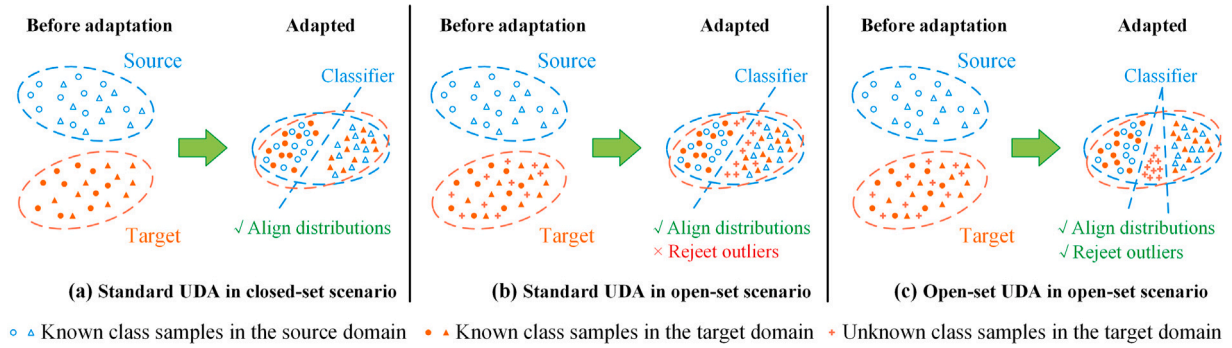


Fig. 1. Contrasting illustrations of closed-set and open-set domain adaptation approaches.

Despite recent advances, deep learning still encounters two fundamental challenges in fault diagnosis. First, conventional deep learning models typically assume that training and testing samples share an identical statistical distribution [14,15]. However, in real industrial environments, variations in operating conditions, noise interference, and equipment degradation often induce substantial distribution shifts, leading to a rapid decline in diagnostic performance as domain discrepancies grow [16,17]. Second, although unsupervised domain adaptation (UDA) methods seek to enable cross-domain knowledge transfer by aligning feature distributions, as illustrated in Fig. 1(a), they remain ineffective under open-set scenarios [18]. Specifically, when the target domain contains fault categories absent in the source domain, conventional UDA methods struggle to correctly identify these unknown samples, as shown in Fig. 1(b), thereby constraining the model's generalization capability.

To address the domain shift problem, two primary strategies have been developed for transferring knowledge from a source domain with abundant annotations to a related but unlabeled target domain. Metric-based domain adaptation methods optimize feature distributions using statistical measures such as MMD [19], CORAL [20], and SWD [21]. For example, Li et al. [22] proposed a weighted joint domain adaptation method for fault diagnosis based on multi-kernel MMD, which minimizes inter-domain discrepancies through a joint distribution discrepancy metric. Shao and Kim [23] introduced an adaptive deep transfer learning framework that integrates CORAL with semantic alignment loss to alleviate domain gaps in cross-domain mechanical fault diagnosis. Wang et al. [24] further designed a joint conditional Wasserstein distance method, aligning source and target distributions under geometric constraints. In contrast, adversarial learning-based approaches achieve domain-invariant representation learning via the adversarial interaction between a feature extractor and a domain discriminator. For instance, Wang et al. [25] proposed a dual-system collaborative adversarial mechanism for unsupervised mechanical fault diagnosis in rotating systems. Similarly, Peng et al. [26] developed an adversarial adaptation framework integrated with MixMatch, where pseudo-label quality is enhanced through confidence thresholds and label mixing strategies, effectively mitigating conditional distribution discrepancies.

The specificity of the open-set UDA problem lies in the coexistence of domain discrepancies and category space differences. Taking a complex gearbox system as an example, typical fault data such as bearing failures can be collected under normal operating conditions. However, under varying working conditions, unforeseen composite fault patterns may emerge. Most existing UDA methods primarily concentrate on aligning domain distributions, yet they lack explicit mechanisms for identifying unknown categories, which often leads to the misclassification of unknown faults into existing categories. Consequently, in cross-domain open-set fault diagnosis, it becomes imperative to introduce mechanisms capable of detecting unknown samples, thereby enabling the model to perceive and recognize previously unseen categories during the adaptation process, as illustrated in Fig. 1(c). To address this, several approaches have been explored. One line of work employs auxiliary binary classifiers with fixed thresholds in adversarial training to determine whether a sample belongs to a known class or represents an outlier [27,28]. Another strategy relies on predefined thresholds applied to measures such as prediction uncertainty, feature similarity, or distance metrics to identify unknown samples [29–31]. In computer vision, the open-set problem was pioneered by Panareda Busto and Gall [32], who employed a linear support vector machine to identify outliers in the target domain. Building on this, Saito et al. [33] proposed a fixed-threshold adversarial learning framework exploiting the low-entropy property of unknown classes to distinguish them from known categories. Pan et al. [34] further advanced this direction with a class-agnostic clustering strategy and mutual information maximization to detect unknown categories. In intelligent fault diagnosis, similar efforts have been made. Zhao and Shen [35] introduced a dual-adversarial learning framework that separates known and unknown faults using adversarial loss and selectively aligns distributions with a weighted domain adversarial loss. Zhang et al. [36] proposed a weighted domain separation framework that establishes decision boundaries between known and unknown categories via an auxiliary binary classifier, eliminating the need for predefined thresholds. More recently, Pan et al. [18] presented a weighted dual-adversarial framework enhanced with supervised contrastive learning, which effectively aligns common fault features while accurately identifying unknown faults in open-set bearing diagnosis under varying working conditions. Despite these advances, current methods still suffer from the risk of feature entanglement between known and unknown categories, particularly when substantial domain shifts occur. This underscores the necessity of developing more robust solutions.

From the above analysis, it is clear that current open-set UDA methods share two main limitations: dependence on fixed-threshold auxiliary classifiers or entropy-based heuristics for unknown detection and inadequate utilization of the intrinsic structure

of unlabeled target data. To address these limitations, this paper proposes a Multi-View Feature Consistency Learning (MFCL) framework for cross-domain open-set fault diagnosis. Specifically, by employing strategies such as source-domain prototype constraints, domain adversarial learning, and multi-view target-sample contrastive learning, the framework achieves effective alignment of feature distributions between the source and target domains and enhances the feature representation capability of unlabeled target samples. An entropy-based threshold-free adversarial separation strategy is adopted to enforce the separation between known and unknown classes. In addition, a novel unknown-aware consistency learning mechanism, constructed using high-quality pseudo-labels generated from target-domain prototypes, further strengthens the discriminability among different classes. The main contributions of this work are summarized as follows:

(1) A novel multi-view feature consistency learning framework is proposed, which integrates source-domain prototype constraints, domain adversarial learning, and multi-view contrastive learning to achieve effective cross-domain feature alignment, while simultaneously enhancing the discriminative representation of unlabeled target-domain samples.

(2) A threshold-free adversarial separation mechanism is designed to address the limitations of existing methods that depend on fixed thresholds or entropy-based heuristics for unknown-class detection, thereby effectively enhancing the separability between known and unknown classes.

(3) An unknown-class-aware consistency learning mechanism is introduced, which leverages target-domain prototypes to construct consistency constraints using high-quality pseudo-labels, thereby further enhancing the discriminative capability of the model across classes in cross-domain scenarios.

The rest of this paper is organized as follows. Section 2 outlines the problem definition along with consistency learning. Section 3 details the proposed approach. Section 4 presents the experimental evaluation, and Section 5 offers concluding remarks.

2. Preliminary theory

2.1. Problem definition

Under dynamically varying operating conditions, frequent changes in speed, load, or other factors may lead to the emergence of novel fault types in the target domain label space, which are referred to as unknown classes. Therefore, under open-set scenarios, the source label set is identified as $C_s = \{1, 2, \dots, C\}$, while the target label set is defined as $C_t = \{1, 2, \dots, C, C+1, C+2, \dots, C'\}$, where $C_s \subset C_t$. The additional labels $C+1, C+2, \dots, C'$ represent unknown classes that are not present in the source domain. To quantify the semantic gap between domains, an openness metric is defined as $\mathcal{O} = 1 - C/C'$, where C is the number of shared (known) classes, and C' is the total number of classes in the target domain. A higher value of $\mathcal{O}$ indicates a greater proportion of unknown classes in the target domain. For generality, all target domain classes that are not included in the source-domain label set are collectively regarded as a single "unknown" class, denoted as C_u. Consequently, the set of target labels is redefined as $C_t = \{1, 2, \dots, C, C_u\}$. The labeled samples associated with the initial working conditions define the source domain $\mathcal{D}_s = \{(x_{s,i}, y_{s,i})\}_{i=1}^{n_s}$, where n_s is the number of labeled samples, $x_{s,i}$ denotes an instance from the source domain, and $y_{s,i}$ is its corresponding class label. The target domain $\mathcal{D}_t = \{x_{t,j}\}_{j=1}^{n_t}$ consists of unlabeled samples under different working conditions, where n_t is the number of target samples and $x_{t,j}$ represents an instance drawn from the target domain.

2.2. Unified dual-stream perturbations consistency

In computer vision, consistency regularization effectively exploits unannotated samples by postulating that a model should generate consistent outputs when subjected to different perturbations of the same image. Under the clustering assumption, this approach performs well when decision boundaries lie in low-density regions of the data distribution. Consistency learning in closed-set UDA focuses on promoting robust representation learning by enforcing prediction consistency on target domain inputs subjected to different augmentations. Earlier consistency regularization approaches aim to align the model's class outputs for weak and strong augmentations of identical samples by minimizing a cross-entropy objective. However, such methods often suffer from limited perturbation space due to their reliance on strong data augmentations alone. UniMatch [37] tackles this problem by integrating perturbations at both the image and feature levels through separate streams, thus enriching the perturbation diversity and promoting more stable feature learning. Specifically, for a mini-batch containing B target domain samples, each sample is first weakly augmented to produce $x_{t,j}^{w}$, from which a perturbed feature representation $x_{t,j}^{fp}$ is extracted. Additionally, two strongly augmented views, $x_{t,j}^{s1}$ and $x_{t,j}^{s2}$, are generated for each sample. In total, this yields $4B$ mixed-augmentation samples, offering diverse perspectives for representation learning, as illustrated in the Fig. 2. Weak augmentations typically involve randomly cropping and flipping images, whereas strong augmentations rely on color transformation strategies [38]. For feature-level perturbations, channel dropout has been demonstrated to be an effective approach [37].

Rather than directly minimizing the discrepancy between the probability distributions induced by different perturbations, UniMatch employs pseudo-labels $\hat{y}_{t,j}^{w}$ generated from weakly augmented samples $x_{t,j}^{w}$ to explicitly guide the consistency learning process of the strongly augmented counterpart. Consequently, the consistency loss $\mathcal{L}_{csl}$ is formulated as follows:

$$\mathcal{L}_{csl} = \frac{1}{B}\sum_{j=1}^{B}\left(\mu\,\mathcal{H}\left(\hat{y}_{t,j}^{w}, p\left(x_{t,j}^{fp}\right)\right) + \frac{\mu}{2}\left(\mathcal{H}\left(\hat{y}_{t,j}^{w}, p\left(x_{t,j}^{s1}\right)\right) + \mathcal{H}\left(\hat{y}_{t,j}^{w}, p\left(x_{t,j}^{s2}\right)\right)\right)\right), \tag{1}$$

where $\mathcal{H}(\cdot)$ denotes the cross-entropy loss function, and μ is a weight coefficient for three loss terms, set to 0.5.

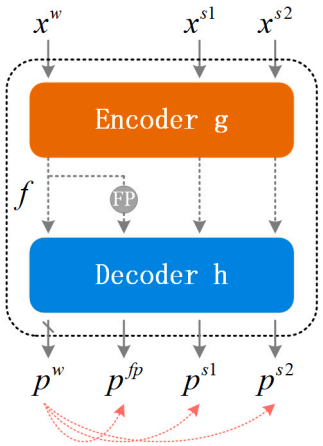


Fig. 2. The UniMatch. The dashed curves denote supervision, and the FP represents feature perturbation.

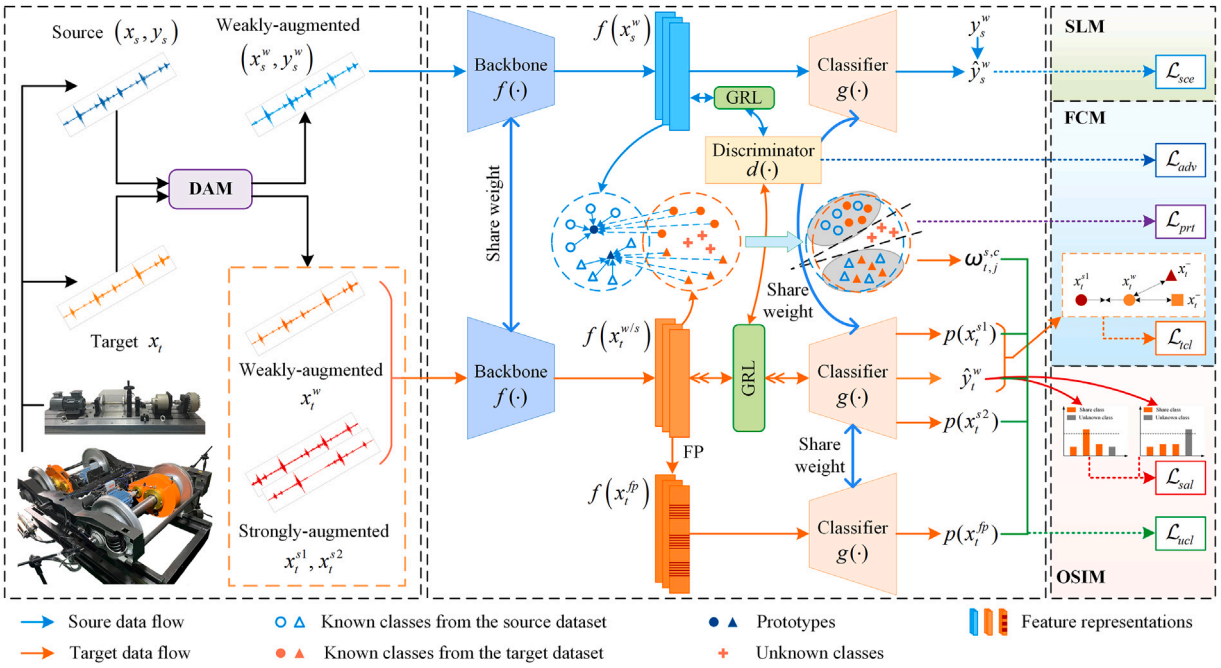


Fig. 3. Architecture of the MFCL framework. Specifically, $f(\cdot)$ indicates the feature extractor parameterized by θ_f, $g(\cdot)$ denotes the label classifier with parameters θ_g, and $d(\cdot)$ represents the domain discriminator with parameters θ_d, while the training objective includes multiple loss components: the source classification loss $\mathcal{L}_{sce}$, the domain adversarial loss $\mathcal{L}_{adv}$, the prototype alignment loss $\mathcal{L}_{prt}$, the multi-view contrastive loss $\mathcal{L}_{tcl}$, the separated adversarial loss $\mathcal{L}_{sal}$, and the unknown-aware consistency loss $\mathcal{L}_{ucl}$.

3. Proposed method

The proposed method for cross-domain open-set fault diagnosis is described in this section, and its overall framework is illustrated in Fig. 3. The open-set UDA task mainly poses two challenges: achieving distribution alignment among known classes and ensuring effective separation between known and unknown classes. To address these problems, a data augmentation module (DAM), a supervised learning module (SLM), a feature constraints module (FCM), and an open-set identification module (OSIM) are designed, which are described in the following sub-sections.

Prior to this, it should be noted that the network architecture of the proposed method consists of a feature extractor $f(\cdot)$ that serves as the backbone for learning meaningful representations from the input data, a domain discriminator $d(\cdot)$ that promotes alignment of feature distributions across domains, and a classifier $g(\cdot)$ that maps the extracted features to class labels, with parameters θ_f, θ_d, and θ_g. The overall network architecture is illustrated in Fig. 4.

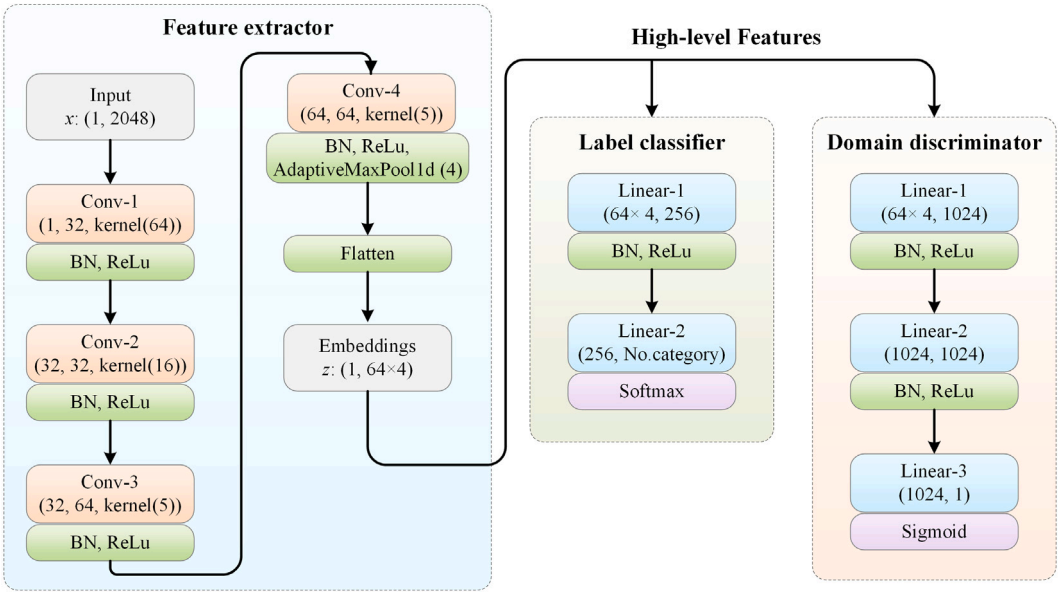
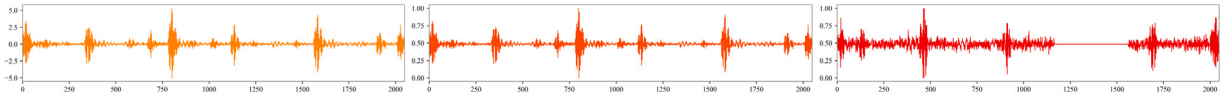


Fig. 4. Network architecture of the proposed approach.



(a) raw signal (b) weak augmentation signal (c) strong augmentation signal

Fig. 5. Comparison of raw signal with weak and strong augmentations.

3.1. Data augmentation module

The DAM is designed to automatically enhance signals from different domains and supports both weak and strong augmentation strategies. Weak augmentation involves normalizing signal data to the range [0, 1]. Strong augmentation, inspired by the random augmentation strategy in [39], applies a combination of transformations with varying intensities, including adding Gaussian SNR noise, time masking, gain adjustment, signal shifting, flipping, fading, and normalization. In this module, only weak augmentation is applied to the source domain data, resulting in weakly augmented samples $x_{s,i}^{w}$. In contrast, both weak and strong augmentations are performed on the target domain data, generating a set of mixed augmented samples: $x_{t,j}^{w}$, $x_{t,j}^{s1}$, and $x_{t,j}^{s2}$, with a corresponding visualization presented in Fig. 5.

3.2. Supervised learning module

Given that the labeled source domain data serve as the only available supervision, the cross-entropy loss $\mathcal{L}_{sce}$ is designed to measure and minimize the discrepancy between the classifier's predictions and the ground-truth labels, which is defined as follows:

$$\mathcal{L}_{sce} = -\frac{1}{n_s}\sum_{i=1}^{n_s}\sum_{c=1}^{C+1} y_{s,i,c}\log\left(\hat{y}_{s,i,c}\right) \tag{2}$$

where $y_{s,i,c}$ represents the true label for class c of sample $x_{s,i}$ from the source domain, and $\hat{y}_{s,i,c}$ represents the corresponding predicted output of the label classifier, i.e., $\hat{y}_{s,i,c} = g(f(x_{s,i}))$. $C+1$ represents the total count of label categories, including C known classes and an additional unknown class. It should be noted that, during training, we utilize the weakly augmented version of the input sample, $x_{s,i}^{w}$, rather than the original input $x_{s,i}$.

3.3. Feature constraints module

Due to feature discrepancies between the source and target domains, accurate recognition in the source domain does not necessarily guarantee reliable diagnosis in the target domain. To address this issue, the FCM is designed to enforce feature consistency. Specifically, prototype constraints and domain adversarial learning are jointly employed to minimize domain discrepancy and align

the source and target feature distributions. Moreover, contrastive learning is introduced to enhance the consistency between strongly and weakly augmented target samples, thereby facilitating the mining of latent discriminative features through data augmentation.

3.3.1. Prototypical constraints on hybrid representations

The motivation for applying prototype constraints is to minimize the distance between learned features and their corresponding class prototypes in the feature space, thereby enhancing intra-class compactness and improving inter-class separability. During domain adaptation, this constraint further guides target-domain features to align with the most probable source-domain prototypes, promoting consistent feature distributions across domains.

The source domain class prototypes are defined as the weighted mean of sample features for each class under label supervision and are initialized as follows:

$$\phi_s^c = \frac{\sum_{\left(x_{s,i}^w, y_{s,i}^w\right)\in D_s} I\left(y_{s,i}^w = c\right) f\left(x_{s,i}^w\right)}{\sum_{\left(x_{s,i}^w, y_{s,i}^w\right)\in D_s} I\left(y_{s,i}^w = c\right)}, \quad c \in C_s \tag{3}$$

where ϕ_s^c represents the feature prototype for the c-th class in the source domain; $y_{s,i}^w$ denotes the true label of the i-th sample; $I(\cdot)$ is the indicator function, which takes the value 1 when $y_{s,i}^w = c$, and 0 otherwise. The term $\sum_{(x_{s,i}^w, y_{s,i}^w)\in D_s} I(y_{s,i}^w = c) f(x_{s,i}^w)$ represents the sum of the feature vectors $f(x_{s,i}^w)$ for the samples labeled as c in the dataset, while $\sum_{(x_{s,i}^w, y_{s,i}^w)\in D_s} I(y_{s,i}^w = c)$ denotes the total number of samples labeled as c. By constructing the set of feature prototypes $\{\phi_s^c\}_{c=1}^C$, representative features for each class can be extracted effectively.

To alleviate the computational cost of iterating over the entire dataset, a mini-batch exponential moving average update strategy is employed:

$$\phi_s^c \leftarrow \lambda_{\text{ema}} \phi_s^c + \left(1 - \lambda_{\text{ema}}\right) \phi_{\text{batch},s}^c \tag{4}$$

where $\phi_{\text{batch},s}^c$ represents the mean feature of class c within the current training batch, and $\lambda_{\text{ema}} = 0.5$ is the momentum coefficient. This strategy dynamically adapts to the evolving trends in the feature space.

To achieve fine-grained alignment between target-domain features and their most similar source-domain prototypes, a prototype-based probability distribution is first defined. Specifically, for a weakly augmented target sample $x_{t,j}^w$, its similarity to each source-domain prototype ϕ_s^c is assessed by the negative Euclidean distance. A softmax function is then applied to convert these similarities into a probability distribution over known classes:

$$p_c\left(x_{t,j}^w\right) = \frac{\exp\left(-\left\| f\left(x_{t,j}^w\right) - \phi_s^c \right\|_2\right)}{\sum_{c'} \exp\left(-\left\| f\left(x_{t,j}^w\right) - \phi_s^{c'} \right\|_2\right)}, \quad c \in C_t \tag{5}$$

where $f(x_{t,j}^w)$ denotes the feature representation of the target sample, and the probability $p_c(x_{t,j}^w)$ indicates the degree of similarity between the target sample and the source prototype of class c.

To encourage target features to align closely with their most similar source prototypes, the prototype loss based on the entropy of the similarity distribution is defined as:

$$\mathcal{L}_{prt} = \frac{1}{n_t} \sum_{j=1}^{n_t} \left(- \sum_{c\in C_t} p_c\left(x_{t,j}^w\right) \log p_c\left(x_{t,j}^w\right) \right) \tag{6}$$

This loss function enhances cross-domain consistency and intra-class compactness by maximizing the similarity between target samples and their corresponding source-domain prototypes.

3.3.2. Adversarial learning for domain alignment

Although applying prototype constraints in open-set UDA minimizes the distance between learned features and class prototypes in the feature space, it also introduces certain limitations. As shown in Eqs. (3) and (5), the unknown category appears only in the target domain and is absent from the source domain. When $c = C_u$, the probability distribution $p_c(x_{t,j}^w)$ differs significantly from the distributions of the known categories. While this characteristic facilitates adversarial separation between known and unknown classes, it also impedes the convergence of the unknown class to the source domain in feature space. Therefore, complementary to achieving fine-grained alignment between domains through prototype constraints, a domain adversarial learning objective is further introduced to globally align the feature distributions of the source and target domains. The objective function for adversarial learning applied to achieve this alignment is formulated as follows:

$$\mathcal{L}_{adv} = -\frac{1}{n_s} \sum_{i=1}^{n_s} \log d\left(f\left(x_{s,i}^w\right)\right) - \frac{1}{n_t} \sum_{j=1}^{n_t} \log\left(1 - d\left(f\left(x_{t,j}^w\right)\right)\right) \tag{7}$$

where $d(\cdot)$ denotes the domain discriminator.

3.3.3. Contrastive learning of multi-view representations

Leveraging the differences between positive and negative samples as self-supervised contrastive information has been shown to effectively learn discriminative representations from unlabeled data [40]. Drawing inspiration from its success on label-free datasets, a discriminative representation learning strategy is applied to the unlabeled multi-view augmented target domain samples, built upon the SimCLR [41] framework.

A positive pair is constructed in the self-supervised contrastive learning scheme by pairing a strongly augmented sample $x_{t,j}^{s1}$ with its weakly augmented counterpart $x_{t,j}^{w}$, both derived from the same unlabeled target sample $x_{t,j}$. Assuming that a training batch contains B target domain samples, the augmentation process results in $2B$ mixed augmented samples. For each positive pair, when using $x_{t,j}^{a}$ as the anchor, the negatives $x_{t,k}^{-}$ are drawn from the set $\{x_{t,k}^{b} \mid k \neq j\}$, where $a \in \{w, s1\}$ and $b \in \{w, s1\}$, $b \neq a$. Consequently, for each positive pair, a total of $2(B-1)$ negatives are utilized across the two symmetric directions. The features of the positive pair $x_{t,j}^{a}$ and $x_{t,j}^{b}$ are projected by the feature extractor $f(\cdot)$ and denoted as $f(x_{t,j}^{a})$ and $f(x_{t,j}^{b})$, respectively. Similarly, negative samples $x_{t,k}^{-}$ are projected as $f(x_{t,k}^{-})$. The objective of contrastive learning is to maximize the similarity between the features of the positive pair $f(x_{t,j}^{a})$ and $f(x_{t,j}^{b})$, while minimizing their similarity with each negative sample $f(x_{t,k}^{-})$, thereby encouraging the learning of more discriminative feature representations. Accordingly, the self-supervised contrastive loss based on multi-view data augmentation for the target domain is defined as follows:

$$\mathcal{L}_{tcl} = -\frac{1}{2B} \sum_{a \in \{w, s1\}} \sum_{j=1}^{B} \log \frac{\exp\left(f(x_{t,j}^{a}) \cdot f(x_{t,j}^{b})/\tau\right)}{\sum_{\substack{k=1 \\ k \neq j}}^{B} \exp\left(f(x_{t,j}^{a}) \cdot f(x_{t,k}^{-})/\tau\right)} \tag{8}$$

where $f(x_{t,j}^{a})$ and $f(x_{t,j}^{b})$ denote the feature projections of the weakly and strongly augmented versions of the same instance $x_{t,j}$, respectively. The anchor $f(x_{t,j}^{a})$ can be either the weak or strong view, with $f(x_{t,j}^{b})$ being the corresponding paired view. The parameter τ is a temperature scaling factor, set to 0.3.

3.4. Open-set identification module

Although domain alignment can be achieved through the aforementioned methods, accurately identifying unknown categories in the target domain remains ineffective under open-set fault diagnosis. To address this issue, a gradient reversal-based separated adversarial learning mechanism is introduced to enable the model to perceive and recognize unknown classes during adaptation, thereby guiding the decision boundary toward a more effective separation between known and unknown samples. Furthermore, by constructing an unknown-aware consistency learning strategy based on high-quality pseudo-labels derived from target-domain prototypes, the discriminability between different classes is further enhanced.

3.4.1. Separated adversarial learning

In open-set UDA, if the model is trained solely by minimizing the cross-entropy loss on labeled source samples, it may overfit to the known classes. As a result, the decision boundary may lack the capacity to effectively distinguish unknown classes, causing the model to misclassify unknown samples as known ones [33]. As noted in [42], due to the absence of semantic supervision for unknown categories, the classifier tends to produce a relatively uniform probability distribution when predicting unknown target samples, resulting in high entropy. In contrast, predictions for known samples generally exhibit lower entropy. Therefore, optimizing the decision boundary by suppressing the prediction entropy of known-class samples while maximizing that of unknown-class samples can guide the model to generate more separable probability distributions for known and unknown classes in the feature space.

To achieve the above objective, a separation adversarial learning loss $\mathcal{L}_{sal}$ is proposed. This loss introduces an adversarial mechanism between the feature extractor and a class discriminator to guide the model in automatically learning the distribution boundary of unknown classes in the feature space. Specifically, a gradient reversal layer (GRL) is employed to enable adversarial optimization during training: the discriminator attempts to determine whether a sample belongs to an unknown class based on its extracted features, while the feature extractor aims to confuse the discriminator. As a result, unknown samples are pushed away from the known class distribution, thereby achieving effective separation between known and unknown classes. The loss function is formulated as follows:

$$\mathcal{L}_{sal} = -\frac{1}{n_t} \sum_{j=1}^{n_t} p\left(\hat{y}_t^{w} = C_u \mid x_{t,j}^{w}\right) \log\left(\sum_{c=1}^{C} p\left(\hat{y}_t^{w} = c \mid x_{t,j}^{w}\right)\right) - \frac{1}{n_t} \sum_{j=1}^{n_t} \left(\sum_{c=1}^{C} p\left(\hat{y}_t^{w} = c \mid x_{t,j}^{w}\right)\right) \log\left(p\left(\hat{y}_t^{w} = C_u \mid x_{t,j}^{w}\right)\right) \tag{9}$$

where $p(\hat{y}_t^{w} = C_u \mid x_{t,j}^{w})$ denotes the probability that the j-th sample belongs to the unknown class C_u, and $\sum_{c=1}^{C} p(\hat{y}_t^{w} = c \mid x_{t,j}^{w})$ denotes the sum of probabilities that the j-th sample belongs to the known classes, with the constraint $p(\hat{y}_t^{w} = C_u \mid x_{t,j}^{w}) + \sum_{c=1}^{C} p(\hat{y}_t^{w} = c \mid x_{t,j}^{w}) = 1$.

3.4.2. Unknown-aware consistency learning

Recent studies have demonstrated that robust models should maintain consistent predictions when subjected to various

transformations of the same input [37,43]. It is important to note that the consistency learning strategy described in Eq. (1) was originally developed for closed-set UDA. For open-set scenarios, it is necessary to first differentiate known samples from unknown ones before applying consistency learning. Accordingly, Eq. (1) is applied separately to the features of known and unknown samples. Specifically, prototype constraints are initially imposed on the target domain to refine the classifier's decision boundary, thereby enabling the generation of more reliable pseudo-labels (distinct from the pseudo-labels $\hat{y}_{t,j}^{w}$ derived from weakly augmented samples $x_{t,j}^{w}$). These high-confidence pseudo-labels are subsequently used to construct a class indicator function $\xi(\cdot)$ for class-aware detection. Finally, under the guidance of the indicator function $\xi(\cdot)$, consistency-regularized self-training is performed separately for samples belonging to known and unknown categories.

Conventional pseudo-label denoising methods typically rely on fixed confidence thresholds to retain high-confidence pseudo-labels while discarding low-confidence ones. However, such approaches are limited in flexibility and often overlook the global distributional structure of the target domain. To more effectively leverage the information latent in unlabeled samples, target-domain prototypes are employed as guidance for pseudo-label denoising. As training progresses toward convergence, the features of unlabeled samples gradually align with their corresponding class centroids, which in turn refines the pseudo-labels for both known and unknown classes and improves the model's awareness of unknown categories. Specifically, the proposed method leverages the feature centroids derived from strongly augmented target samples as class prototypes, which are iteratively employed to update pseudo-labels and adaptively mitigate the influence of noisy predictions. The classification weights are obtained by calculating the distances between the target prototypes $\{\phi_t^c\}_{c=1}^{C_u}$ and the features of the target domain samples, as elaborated below:

$$\phi_t^{s,c} = \frac{\sum_{x_{t,i}^s \in \mathcal{D}_t} I\left(\hat{y}_{t,j}^{w} = c\right) f\left(x_{t,j}^{s}\right)}{\sum_{x_{t,j}^s \in \mathcal{D}_t} I\left(\hat{y}_{t,j}^{w} = c\right)}, \quad c \in C_t,\ s \in \{s1, s2\} \tag{10}$$

$$\omega_{t,j}^{s,c} = \frac{\exp\left(-\left\| f\left(x_{t,j}^{w}\right) - \phi_t^{s,c} \right\|_2\right)}{\sum_{c'} \exp\left(-\left\| f\left(x_{t,j}^{w}\right) - \phi_t^{s,c'} \right\|_2\right)}, \quad c \in C_t,\ s \in \{s1, s2\} \tag{11}$$

where $\phi_t^{s,c}$ denotes the feature prototype of class c in the strongly augmented samples, and $\omega_{t,j}^{s,c}$ represents the weight indicating the likelihood of the sample $x_{t,j}^{w}$ belonging to class c.

As observed from Eq. (11), samples that are closer to the feature centroid are regarded as more reliable, whereas those that are farther from the prototype are likely to be outliers. Accordingly, the classification probabilities are progressively updated based on this principle. By progressively updating the weights using newly acquired knowledge, the model is able to generate more reliable pseudo-labels:

$$M_{t,j}^{w,c} = \begin{cases} 1, & \text{if } c = \arg\max_{c'} \ \omega_{t,j}^{s1,c'} \cdot p^{c'}\left(x_{t,j}^{w}\right) \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

$$M_{t,j}^{s1,c} = \begin{cases} 1, & \text{if } c = \arg\max_{c'} \ \omega_{t,j}^{s2,c'} \cdot p^{c'}\left(x_{t,j}^{s1}\right) \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

where $M_{t,j}^{w,c}$ denotes the mask indicating that the sample $x_{t,j}^{w}$ belongs to class c, and $p^{c'}(x_{t,j}^{w})$ represents the predicted probability of $x_{t,j}^{w}$ belonging to class c'.

Accordingly, an indicator function $\xi(\cdot)$ can be defined to identify whether a given target sample belongs to a known or unknown class:

$$\xi\left(M_{t,j}^{a,c}\right) = \begin{cases} 1, & \text{if } M_{t,j}^{a,c} \in \{1, 2, \ldots, C\} \\ 0, & \text{otherwise,} \end{cases} \quad a \in \{w, s1\} \tag{14}$$

Therefore, the unknown-aware consistency loss based on the indicator function $\xi(\cdot)$ can be defined as follows:

$$\begin{aligned} \mathcal{L}_{ucl} = & \sum_{a \in \{w, s1\}} \frac{1}{\gamma_1 B} \sum_{i=1}^{B} \xi\left(M_{t,j}^{a,c}\right) \cdot \left[\mu \mathcal{H}\left(\hat{y}_{t,j}^{w}, p\left(x_{t,j}^{fp}\right)\right) + \frac{\mu}{2}\left(\mathcal{H}\left(\hat{y}_{t,j}^{w}, p\left(x_{t,j}^{s1}\right)\right) + \mathcal{H}\left(\hat{y}_{t,j}^{w}, p\left(x_{t,j}^{s2}\right)\right)\right)\right] \\ & + \sum_{a \in \{w, s1\}} \frac{1}{\gamma_2 B} \sum_{i=1}^{B} \left(1 - \xi\left(M_{t,j}^{a,c}\right)\right) \cdot \left[\mu \mathcal{H}\left(C_u, p\left(x_{t,j}^{fp}\right)\right) + \frac{\mu}{2}\left(\mathcal{H}\left(C_u, p\left(x_{t,j}^{s1}\right)\right) + \mathcal{H}\left(C_u, p\left(x_{t,j}^{s2}\right)\right)\right)\right] \end{aligned} \tag{15}$$

where γ_1 and γ_2 denote the respective ratios of known and unknown instances included within each mini-batch, $a \in \{w, s_1\}$ represents the two different augmented sample sets, and $\hat{y}_{t,j}^{w} \in \{1, 2, \ldots, C\}$ indicates the predicted class label for the j-th target sample in the weakly augmented view.

Algorithm 1 The implementation procedure of MFCL.

Require: Source domain dataset $\mathcal{D}_s$, target domain dataset $\mathcal{D}_t$, feature extractor $f(\cdot)$, classifier $g(\cdot)$, discriminator $d(\cdot)$ with parameters θ_f, θ_g, θ_d, learning rate η, temperature τ, momentum λ_{ema}, and weights λ_{sal}, λ_{ucl}, λ_{prt}, λ_{adv}, λ_{tcl}.
Ensure: Well-trained $f(\cdot)$, $g(\cdot)$, and $d(\cdot)$.

Initialize prototypes: $\phi_s^c \leftarrow (y_{s,i}^w, f(x_{s,i}^w))$, $\phi_t^{s,c} \leftarrow (\hat{y}_{t,j}^w, f(x_{t,j}^s))$
for $k = 0$ **to** $K-1$ **do**
  if $k < 5$ **then**
    Warm-up phase, update parameters θ_f and θ_g:

$$\theta_f \leftarrow \theta_f - \eta\left(\frac{\partial \mathcal{L}_{cls}}{\partial \theta_f} + \lambda_{tcl}\frac{\partial \mathcal{L}_{tcl}}{\partial \theta_f}\right)$$

$$\theta_g \leftarrow \theta_g - \eta\frac{\partial \mathcal{L}_{cls}}{\partial \theta_g}$$

  else
    Main phase, update prototypes:

$$\phi_s^c \leftarrow \lambda_{ema}\phi_s^c + \left(1-\lambda_{ema}\right)\phi_{batch,s}^c$$

$$\phi_t^{s,c} \leftarrow \lambda_{ema}\phi_t^{s,c} + \left(1-\lambda_{ema}\right)\phi_{batch,t}^{s,c}$$

    Compute high-confidence pseudo labels:

$$\omega_{t,j}^{s,c} \leftarrow \left(f(x_{t,j}^w), \phi_t^{s,c}\right)$$

$$M_{t,j}^{w,c} \leftarrow \omega_{t,j}^{s1,c'} \cdot p^{c'}(x_{t,j}^w) \quad \text{and} \quad M_{t,j}^{s1,c} \leftarrow \omega_{t,j}^{s2,c'} \cdot p^{c'}(x_{t,j}^{s1})$$

    Apply class identification mechanism: $\xi(M_{t,j}^{w,c}) \leftarrow M_{t,j}^{w,c}$ and $\xi(M_{t,j}^{s1,c}) \leftarrow M_{t,j}^{s1,c}$
    Update parameters θ_f and θ_g and θ_d for $f(\cdot)$, $g(\cdot)$ and $d(\cdot)$:

$$\theta_f \leftarrow \theta_f - \eta\left(\frac{\partial \mathcal{L}_{cls}}{\partial \theta_f} - \lambda_{sal}\frac{\partial \mathcal{L}_{sal}}{\partial \theta_f} + \lambda_{ucl}\frac{\partial \mathcal{L}_{ucl}}{\partial \theta_f} + \lambda_{prt}\frac{\partial \mathcal{L}_{prt}}{\partial \theta_f} - \lambda_{adv}\frac{\partial \mathcal{L}_{adv}}{\partial \theta_f} + \lambda_{tcl}\frac{\partial \mathcal{L}_{tcl}}{\partial \theta_f}\right)$$

$$\theta_g \leftarrow \theta_g - \eta\left(\frac{\partial \mathcal{L}_{cls}}{\partial \theta_g} - \lambda_{sal}\frac{\partial \mathcal{L}_{sal}}{\partial \theta_g} + \lambda_{ucl}\frac{\partial \mathcal{L}_{ucl}}{\partial \theta_g}\right)$$

$$\theta_d \leftarrow \theta_d - \eta\lambda_{adv}\frac{\partial \mathcal{L}_{adv}}{\partial \theta_d}$$

  end if
end for

3.5. Optimization objective

Incorporating the aforementioned loss terms, the overall objective function of MFCL is defined as follows:

$$\mathcal{L}_{total}(\theta_f, \theta_g, \theta_d) = \mathcal{L}_{sce} - \lambda_{sal}\mathcal{L}_{sal} + \lambda_{ucl}\mathcal{L}_{ucl} + \lambda_{prt}\mathcal{L}_{prt} - \lambda_{adv}\mathcal{L}_{adv} + \lambda_{tcl}\mathcal{L}_{tcl} \tag{16}$$

where λ_{sal}, λ_{ucl}, λ_{prt}, λ_{adv}, and λ_{tcl} are the weight coefficients for $\mathcal{L}_{sal}$, $\mathcal{L}_{ucl}$, $\mathcal{L}_{prt}$, $\mathcal{L}_{adv}$, and $\mathcal{L}_{tcl}$, respectively. The objective function is optimized based on the parameters θ_f, θ_g, and θ_d. Notably, entropy separation is implicitly performed by maximizing $\mathcal{L}_{sal}$ using the GRL. Algorithm 1 summarizes the training process.

During inference, the target domain classification results are obtained by passing only the weakly augmented test samples through the well-trained feature extractor $f(\cdot)$, followed by the classifier $g(\cdot)$.

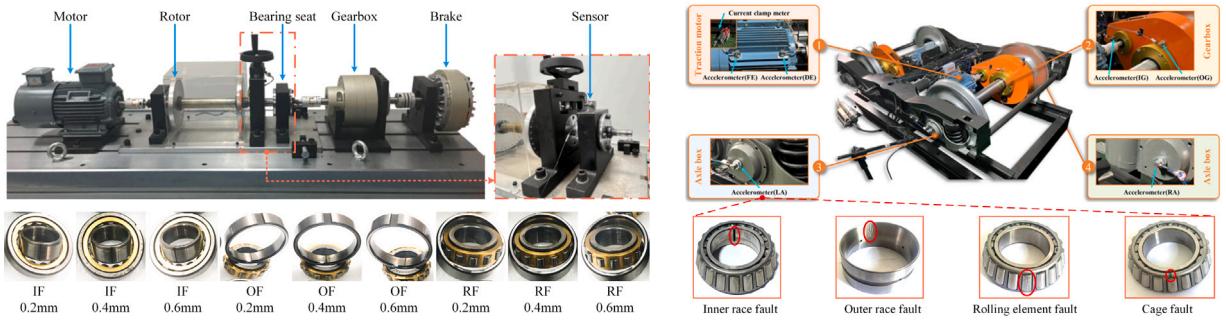
4. Experimental evaluation

4.1. Dataset description

(1) SDUST.[1] The first bearing dataset originates from Shandong University of Science and Technology [44]. Vibration signals were collected from an accelerometer installed at the drive end of the motor, under two different motor load conditions: 2000 rpm with a 20 N load and 2000 rpm with a 40 N load. The dataset includes three severity levels for each fault type, yielding ten distinct bearing health states: normal condition (NC), inner ring faults (IF0.2, IF0.4, IF0.6), outer ring faults (OF0.2, OF0.4, OF0.6), and rolling element faults (RF0.2, RF0.4, RF0.6).

(2) BJTU-RAO.[2] The second bearing dataset is sourced from the bogie fault diagnosis platform at Beijing Jiaotong University [45]. Two representative operating conditions are selected: 20 Hz with a 0 kN load and 60 Hz with a −10 kN load (where the negative sign indicates that the lateral load is applied toward the gearbox side of the tested drivetrain). Vibration signals were recorded via an accelerometer installed on the left side of the axle box. The dataset covers five typical bearing health conditions: normal condition (NC), inner race fault (IF), outer race fault (OF), rolling element fault (RF), and cage fault (CF).

[1] https://github.com/JRWang-SDUST/SDUST-Dataset.git
[2] https://pan.baidu.com/s/1S5F8URapHPArX2mV6FzDfA?pwd=9t74



(a) SDUST (b) BJTU-RAO

Fig. 6. Test rigs of the SDUST and BJTU-RAO datasets.

Table 1
Details of the SDUST and BJTU-RAO datasets.

Dataset	Category Labels	0	1	2	3	4	5	6	7	8	9
SDUST	Fault location	NC	IF	IF	IF	OF	OF	OF	RF	RF	RF
	Fault size (mm)	0	0.2	0.4	0.6	0.2	0.4	0.6	0.2	0.4	0.6
BJTU-RAO	Fault location	NC	IF	OF	RF	CF					

Table 2
Transfer task definitions in the SDUST and BJTU-RAO datasets.

SDUST				BJTU-RAO			
Task	Transfer (load)	Source classes	$\mathcal{O}$	Task	Transfer (speed & load)	Source classes	$\mathcal{O}$
S1	2000rpm_20N → 2000rpm_40N	0,1,2,3,4,5,6,7,8	0.1	B1	20 Hz_0kN → 60 Hz_-10kN	0,1,2,3	0.2
S2		0,1,2,3,4,5,6,7	0.2	B2		0,1,2	0.4
S3		0,1,2,3,4,5,6	0.3	B3		0,1	0.6
S4		0,1,2,3,4,5	0.4	B4		0,2	0.6
S5	2000rpm_40N → 2000rpm_20N	0,1,2,3,4	0.5	B5	60 Hz_-10kN → 20 Hz_0kN	0,2,3,4	0.2
S6		0,1,2,3	0.6	B6		0,3,4	0.4
S7		0,1,2	0.7	B7		0,3	0.6
S8		0,1	0.8	B8		0,4	0.6

Fig. 6 illustrates the test platforms used for the two bearing datasets, while Table 1 summarizes the corresponding machine health states. For each health state, 320 training samples and 80 testing samples were collected, with each sample consisting of 2048 data points. To evaluate the effectiveness and robustness of the proposed method under varying degrees of openness, a total of 16 cross-domain fault diagnosis tasks with different openness levels were designed. All tasks were randomly generated, and detailed configuration information is provided in Table 2.

4.2. Comparison methods and implementation details

(1) Method comparison. To evaluate the effectiveness and superiority of the proposed MFCL method, several representative domain adaptation approaches are selected for comparison. The comparison involves one closed-set UDA method, DANN [46], and seven open-set UDA methods, including OSBP [33], DAOSDA [35], ANNA [47], MAOSDAN [28], RRDA+shot [48], COSDA [49], and PUMCL [50]. All comparison methods adopt the same backbone network and classifier structure as shown in Fig. 4 to ensure fair comparisons across experiments.

(2) Parameter settings. The hyperparameters for all comparison methods are configured based on the corresponding literature and experimental requirements to ensure representative performance. Model training is conducted using the PyTorch framework on a computing platform equipped with an NVIDIA RTX 4070 GPU (12 GB VRAM). Each task is repeated five times, and the average accuracy along with the standard deviation is reported to mitigate the effects of randomness. To ensure fair comparisons, all methods adopt the same training strategy. Specifically, stochastic gradient descent (SGD) with a momentum factor of 0.9 is employed as the optimizer. A learning rate of 0.001 is adopted, according to the annealing strategy suggested by [51]. The model is trained for 75 epochs with a batch size of 32. The weighting coefficients in the total loss are set to $\lambda_{sal} = 1.0$, $\lambda_{ucl} = 1.0$, $\lambda_{prt} = 0.5$, $\lambda_{adv} = 0.5$, and $\lambda_{tcl} = 0.5$. The code of MFCL is available at https://github.com/ChnGuiping/MFCL.

4.3. Evaluation metrics

To follow established conventions in the literature [52,53], four evaluation metrics are adopted to comprehensively assess the

performance of the proposed approach. These metrics include the normalized accuracy for overall samples (OS), the accuracy for known class samples (KNO), the accuracy for unknown class samples (UNK), and the harmonic mean (HOS) of KNO and UNK. The definitions of these metrics can be formulated as follows:

$$OS = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \tag{17}$$

$$KNO = \frac{1}{C} \sum_{c=1}^{C} \frac{TP_c}{TP_c + FP_c} \times 100\% \tag{18}$$

$$UNK = \frac{TP_{C_u}}{TP_{C_u} + FP_{C_u}} \times 100\% \tag{19}$$

$$HOS = 2 \times \frac{KNO \times UNK}{KNO + UNK} \times 100\% \tag{20}$$

where TP indicates the counts of true positives, FP are false positives, TN reflects true negatives, and FN represents false negatives, TP_c represents the count of samples correctly predicted as class c, and FP_c refers to the number of instances incorrectly classified as class c despite belonging to other classes.

In open-set UDA evaluation, although the conventional OS metric reflects the overall performance on both known and unknown classes, it suffers from performance distortion when the number of known (or unknown) classes is relatively large, as the evaluation weight of the unknown (or known) classes becomes diluted. Compared to the OS metric, the HOS metric more effectively quantifies an algorithm's ability to recognize both known and unknown classes, thereby providing a more comprehensive performance evaluation.

4.4. Experimental findings and analysis

4.4.1. Quantitative analysis

The evaluation outcomes of all approaches on the 16 open-set cross-domain tasks are present in Tables 3 and 4. As evidenced by the tables, due to the domain shift effects, the source-only approach (i.e., without any domain adaptation strategy) performs the poorest and fails to correctly recognize both known and unknown categories.

In cross-domain open-set fault diagnosis tasks, the model needs to balance effective identification of unknown classes with accurate classification of known classes to achieve optimal performance. In contrast to open-set UDA methods, closed-set UDA approaches perform excellently on known classes, typically exhibiting high KNO scores. However, owing to the absence of prior knowledge about unknown categories, these methods usually only achieve domain alignment between the source and target domains, failing to correctly recognize unknown samples in the target domain. This limitation often leads to a decline in overall performance, as reflected by low OS values.

Moreover, open-set UDA methods may suffer from decision boundary shifts caused by feature overlap introduced by unknown classes. These shifts, which attempt to harmonize the classification of known and unknown instances, can lead to a decline in the KNO metric. As shown in Tables 3 and 4, smaller openness values (e.g., S1, S2, B1, and B5) lead to poorer KNO performance. This is primarily because, during the process of adversarially separating known and unknown categories, the number of unknown samples is significantly smaller than that of known ones, prompting the model to adjust the decision boundary in favor of recognizing unknown instances. While this facilitates the identification of unknown classes, it often results in confusion among the known class features. In contrast, when the openness is larger (e.g., S6, S7, S8, B3, and B4), where unknown samples outnumber the known ones, the KNO metric of open-set UDA methods approaches that of standard UDA, while the UNK metric tends to decline. Nevertheless, regardless of the openness level, open-set UDA methods consistently demonstrate improved capability in identifying unknown classes. As a result, they achieve superior performance compared to closed-set UDA methods on comprehensive metrics such as OS and HOS.

Specifically, on the SDUST dataset (Table 3), the proposed MFCL method achieves an average OS of 92.88% and an average HOS of 91.83%, outperforming the second-best open-set UDA method by 4.63% and 4.99%, respectively. It obtains the best OS and HOS scores in 5 out of 8 tasks (S2, S3, S6, S7, and S8). The method also achieves the highest average UNK (90.58%) and KNO (94.28%) scores among all evaluated open-set UDA approaches. On the BJTU-RAO dataset (Table 4), the proposed method yields an average OS of 96.16% and an average HOS of 96.51%, surpassing the second-best method by 1.52% and 2.33% and outperforming the third-best by 5.74% and 8.64%, respectively. It achieves the best OS and HOS in 5 of the 8 tasks (B2, B3, B6, B7, and B8). Furthermore, its average KNO and UNK scores reach 96.18% and 97.50%, respectively, both exceeding those of other open-set UDA methods. Notably, while maintaining high recognition accuracy for unknown samples, the proposed method achieves a KNO performance on the BJTU-RAO dataset that closely approximates that of standard UDA. These results indicate that the proposed MFCL method effectively recognizes unknown machine health states in open-set cross-domain conditions and consistently achieves superior overall performance compared to alternative approaches. It also achieves a favorable balance between known and unknown class recognition. This superior performance is attributed to the support of self-supervised contrastive learning and class-aware consistency learning (as evidenced in Section 4.5.1: *Effectiveness analysis of key modules*), enabling stable and promising results across all datasets.

Table 3
Classification accuracy (%) on open-set tasks of the SDUST dataset. The optimal and sub-optimal results are highlighted in red and blue, respectively.

Method		Closed-set UDA		Open-set UDA							
		Source-only	DANN	OSBP	DAOSDA	ANNA	MAOSDAN	RRDA+shot	COSDA	PUMCL	MFCL
S1	OS	74.53 ± 0.69	88.47 ± 2.10	76.92 ± 2.12	72.58 ± 3.08	84.50 ± 4.09	72.60 ± 0.97	82.03 ± 1.18	71.85 ± 5.05	77.78 ± 2.87	83.33 ± 1.29
	KNO	82.81 ± 0.77	98.31 ± 2.34	78.81 ± 1.93	70.83 ± 3.27	83.19 ± 4.10	71.08 ± 1.84	87.17 ± 1.64	70.00 ± 5.42	76.44 ± 2.62	81.50 ± 1.47
	UNK	/	/	60.00 ± 8.97	88.25 ± 5.05	96.25 ± 4.33	86.25 ± 8.34	35.75 ± 7.94	88.50 ± 7.78	89.75 ± 10.32	99.75 ± 0.56
	HOS	/	/	67.85 ± 5.92	78.52 ± 3.16	89.24 ± 4.12	77.71 ± 2.41	50.24 ± 7.92	77.99 ± 4.80	82.37 ± 5.10	89.70 ± 0.78
S2	OS	77.00 ± 1.64	87.25 ± 1.13	86.61 ± 1.09	80.64 ± 1.45	86.28 ± 3.72	77.50 ± 3.59	82.15 ± 3.19	76.89 ± 3.36	87.13 ± 1.40	88.64 ± 2.14
	KNO	86.62 ± 1.84	98.16 ± 1.27	86.72 ± 0.88	79.72 ± 1.18	86.25 ± 4.55	76.47 ± 2.52	87.84 ± 3.73	75.16 ± 3.43	86.19 ± 1.48	87.66 ± 2.27
	UNK	/	/	85.75 ± 3.38	88.00 ± 4.87	86.50 ± 8.77	85.75 ± 12.37	36.62 ± 7.97	90.75 ± 3.91	94.62 ± 2.75	96.50 ± 3.38
	HOS	/	/	86.21 ± 2.05	83.62 ± 2.70	86.06 ± 4.13	80.61 ± 7.28	51.25 ± 7.52	82.20 ± 3.37	90.19 ± 1.58	91.84 ± 2.22
S3	OS	78.25 ± 0.68	87.50 ± 0.00	95.93 ± 1.51	89.29 ± 3.15	92.84 ± 1.21	89.67 ± 4.06	83.74 ± 1.52	82.23 ± 1.32	98.33 ± 1.04	99.71 ± 0.30
	KNO	89.43 ± 0.77	100.00 ± 0.00	96.68 ± 2.08	89.11 ± 3.40	93.64 ± 1.86	89.36 ± 3.93	90.75 ± 1.62	80.43 ± 1.55	98.21 ± 1.02	99.68 ± 0.32
	UNK	/	/	90.67 ± 3.07	90.58 ± 5.50	87.25 ± 8.35	91.83 ± 6.93	34.67 ± 2.64	94.83 ± 1.58	99.17 ± 1.21	99.92 ± 0.19
	HOS	/	/	93.52 ± 0.95	89.76 ± 3.42	90.11 ± 4.27	90.51 ± 5.00	50.12 ± 2.82	87.02 ± 0.97	98.69 ± 1.11	99.80 ± 0.23
S4	OS	76.25 ± 0.77	83.93 ± 3.89	93.95 ± 7.98	91.65 ± 2.36	94.12 ± 2.28	96.38 ± 2.72	67.07 ± 6.06	87.71 ± 1.08	95.29 ± 3.92	96.29 ± 8.11
	KNO	88.96 ± 0.90	97.92 ± 4.54	95.42 ± 7.32	92.92 ± 2.56	95.75 ± 2.79	96.21 ± 3.24	74.29 ± 6.83	86.75 ± 1.26	95.29 ± 3.57	96.58 ± 7.41
	UNK	/	/	85.12 ± 12.61	84.06 ± 2.90	84.37 ± 5.88	97.44 ± 1.02	23.75 ± 2.23	93.44 ± 1.04	95.25 ± 10.10	94.50 ± 12.30
	HOS	/	/	89.85 ± 10.29	88.24 ± 2.27	89.58 ± 3.13	96.79 ± 1.55	35.96 ± 3.14	89.96 ± 0.79	95.08 ± 6.35	95.47 ± 10.02
S5	OS	80.62 ± 0.53	83.04 ± 0.11	94.27 ± 6.32	93.94 ± 0.78	91.73 ± 3.30	94.71 ± 2.66	66.76 ± 2.75	91.01 ± 0.93	93.57 ± 3.79	97.34 ± 3.07
	KNO	96.75 ± 0.64	99.65 ± 0.14	94.70 ± 6.02	93.00 ± 0.98	96.40 ± 3.34	95.10 ± 0.45	76.05 ± 3.15	89.75 ± 1.16	94.65 ± 2.54	98.25 ± 2.28
	UNK	/	/	92.15 ± 8.05	98.65 ± 1.04	68.40 ± 11.85	92.75 ± 14.82	20.30 ± 0.80	97.30 ± 1.05	88.15 ± 10.89	92.80 ± 9.86
	HOS	/	/	93.39 ± 7.02	95.74 ± 0.60	79.51 ± 8.42	93.38 ± 8.62	32.04 ± 1.26	93.37 ± 0.67	91.10 ± 6.81	95.26 ± 5.91
S6	OS	76.45 ± 0.78	79.85 ± 0.14	80.91 ± 5.16	83.38 ± 11.00	82.49 ± 3.82	87.67 ± 9.60	51.34 ± 2.84	86.70 ± 11.03	88.41 ± 6.72	93.69 ± 4.81
	KNO	95.56 ± 0.97	99.81 ± 0.17	83.94 ± 5.11	82.06 ± 11.43	89.31 ± 5.76	90.50 ± 7.80	59.25 ± 3.97	87.44 ± 11.95	90.44 ± 8.24	96.31 ± 3.53
	UNK	/	/	68.79 ± 8.83	88.62 ± 9.76	55.21 ± 6.84	76.33 ± 25.33	19.71 ± 1.90	83.75 ± 7.34	80.29 ± 7.54	83.21 ± 10.99
	HOS	/	/	75.41 ± 6.81	85.17 ± 10.54	67.84 ± 4.62	81.18 ± 17.55	29.46 ± 1.60	85.51 ± 9.57	84.75 ± 5.83	89.10 ± 7.52
S7	OS	74.50 ± 0.17	75.00 ± 0.00	84.05 ± 1.52	85.40 ± 5.50	83.86 ± 2.54	86.96 ± 4.75	65.88 ± 4.32	68.66 ± 4.38	80.75 ± 11.53	88.68 ± 9.12
	KNO	99.33 ± 0.23	100.00 ± 0.00	95.67 ± 1.52	90.00 ± 4.61	94.42 ± 3.34	96.08 ± 1.20	79.42 ± 6.16	68.17 ± 5.85	83.17 ± 15.59	94.50 ± 12.30
	UNK	/	/	49.21 ± 2.08	71.61 ± 10.54	52.18 ± 3.23	59.57 ± 18.39	25.25 ± 3.26	70.14 ± 3.07	73.50 ± 0.97	71.21 ± 0.44
	HOS	/	/	64.98 ± 2.05	79.51 ± 7.98	67.14 ± 2.76	71.97 ± 16.72	38.14 ± 3.36	69.00 ± 3.25	77.44 ± 6.46	80.89 ± 4.84
S8	OS	66.67 ± 0.00	66.67 ± 0.00	73.42 ± 0.66	89.17 ± 3.03	77.91 ± 1.25	78.32 ± 2.20	82.01 ± 0.20	84.10 ± 4.70	84.78 ± 2.11	95.41 ± 3.06
	KNO	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	99.88 ± 0.28	100.00 ± 0.00	100.00 ± 0.00	98.88 ± 0.93	92.00 ± 6.41	95.62 ± 8.41	99.75 ± 0.56
	UNK	/	/	20.25 ± 1.98	67.75 ± 8.58	33.72 ± 3.75	34.97 ± 6.60	48.28 ± 1.66	68.31 ± 2.02	63.09 ± 10.61	86.72 ± 8.90
	HOS	/	/	33.64 ± 2.78	80.48 ± 6.44	50.34 ± 4.14	51.54 ± 6.99	64.86 ± 1.30	78.35 ± 3.31	75.09 ± 3.60	92.59 ± 5.11
Avg	OS	75.53 ± 4.10	81.46 ± 7.47	85.76 ± 8.46	85.76 ± 6.88	86.72 ± 5.69	85.48 ± 8.53	72.62 ± 11.68	81.14 ± 7.94	88.25 ± 7.16	92.88 ± 5.50
	KNO	92.43 ± 6.33	99.23 ± 0.93	91.49 ± 7.40	87.19 ± 9.17	92.37 ± 5.64	89.35 ± 10.28	81.71 ± 12.19	81.21 ± 9.20	90.00 ± 7.50	94.28 ± 6.45
	UNK	/	/	68.99 ± 25.06	84.69 ± 10.19	70.49 ± 21.77	78.11 ± 21.07	30.54 ± 9.94	85.88 ± 11.08	85.48 ± 12.33	90.58 ± 9.79
	HOS	/	/	75.61 ± 20.31	85.13 ± 5.89	77.48 ± 14.43	80.46 ± 14.40	44.01 ± 12.06	82.93 ± 7.76	86.84 ± 8.35	91.83 ± 5.61

Table 4
Classification accuracy (%) on open-set tasks of the BJTU-RAO dataset. The optimal and sub-optimal results are highlighted in red and blue, respectively.

Method		Closed-set UDA		Open-set UDA							
		Source-only	DANN	OSBP	DAOSDA	ANNA	MAOSDAN	RRDA+shot	COSDA	PUMCL	MFCL
B1	OS	76.10 ± 2.36	80.00 ± 0.00	98.45 ± 0.74	97.45 ± 1.36	98.65 ± 0.29	89.00 ± 16.27	77.75 ± 2.78	81.65 ± 2.59	94.15 ± 3.62	97.30 ± 6.04
	KNO	95.12 ± 2.94	100.00 ± 0.00	99.31 ± 0.51	96.88 ± 1.67	99.94 ± 0.14	88.69 ± 15.14	89.69 ± 4.08	81.69 ± 2.78	92.81 ± 4.45	96.62 ± 7.55
	UNK	/	/	95.00 ± 1.77	99.75 ± 0.56	93.50 ± 1.63	90.25 ± 21.11	30.00 ± 4.92	81.50 ± 2.71	99.50 ± 0.68	100.00 ± 0.00
	HOS	/	/	97.10 ± 1.14	98.29 ± 0.95	96.61 ± 0.84	89.33 ± 18.19	44.71 ± 5.14	81.58 ± 2.47	96.00 ± 2.56	98.16 ± 4.12
B2	OS	71.94 ± 2.09	65.25 ± 8.03	76.75 ± 21.38	85.66 ± 16.39	98.81 ± 0.39	90.84 ± 11.30	71.84 ± 7.95	73.59 ± 10.90	99.81 ± 0.17	100.00 ± 0.00
	KNO	95.92 ± 2.79	87.00 ± 10.71	85.75 ± 16.92	89.08 ± 13.45	99.83 ± 0.37	92.08 ± 9.62	77.42 ± 5.55	76.25 ± 9.35	99.75 ± 0.23	100.00 ± 0.00
	UNK	/	/	49.75 ± 36.42	75.38 ± 27.99	95.75 ± 1.03	87.12 ± 17.35	55.12 ± 17.54	65.62 ± 17.52	100.00 ± 0.00	100.00 ± 0.00
	HOS	/	/	58.96 ± 34.64	80.42 ± 22.24	97.75 ± 0.57	89.29 ± 13.57	63.34 ± 14.57	70.15 ± 13.18	99.87 ± 0.11	100.00 ± 0.00
B3	OS	66.67 ± 0.00	66.67 ± 0.00	83.11 ± 4.48	86.31 ± 9.96	82.64 ± 2.89	89.14 ± 3.27	90.33 ± 3.26	97.50 ± 0.33	93.08 ± 4.46	98.83 ± 1.10
	KNO	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	89.62 ± 9.52	100.00 ± 0.00	98.25 ± 1.43	97.38 ± 2.36	98.75 ± 0.44	100.00 ± 0.00	100.00 ± 0.00
	UNK	/	/	49.33 ± 13.43	79.67 ± 11.33	47.92 ± 8.66	70.92 ± 11.37	76.25 ± 6.65	95.00 ± 0.66	79.25 ± 13.38	96.50 ± 3.29
	HOS	/	/	65.24 ± 11.53	84.29 ± 10.44	64.42 ± 7.91	81.90 ± 7.57	85.43 ± 4.55	96.84 ± 0.36	87.92 ± 8.42	98.20 ± 1.71
B4	OS	66.67 ± 0.00	66.67 ± 0.00	80.22 ± 1.55	95.06 ± 2.56	84.89 ± 3.40	87.72 ± 3.25	89.31 ± 3.14	96.97 ± 0.15	94.39 ± 3.38	95.61 ± 4.10
	KNO	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	99.75 ± 0.34	100.00 ± 0.00	95.75 ± 3.55	95.31 ± 5.79	98.38 ± 0.34	100.00 ± 0.00	100.00 ± 0.00
	UNK	/	/	40.67 ± 4.64	85.67 ± 7.85	54.67 ± 10.19	71.67 ± 9.62	77.29 ± 7.43	94.17 ± 0.51	83.17 ± 10.15	86.83 ± 12.31
	HOS	/	/	57.70 ± 4.52	92.02 ± 4.45	70.25 ± 8.31	81.60 ± 6.06	85.01 ± 3.75	96.22 ± 0.18	90.52 ± 6.50	92.55 ± 7.68
B5	OS	78.35 ± 0.80	78.35 ± 0.38	88.65 ± 3.87	91.15 ± 1.42	88.15 ± 2.32	84.70 ± 5.07	73.40 ± 2.00	69.10 ± 1.45	82.20 ± 3.20	80.10 ± 1.46
	KNO	97.94 ± 1.00	97.94 ± 0.47	92.25 ± 4.85	91.44 ± 1.03	93.81 ± 1.85	82.94 ± 5.90	81.00 ± 2.84	66.37 ± 3.19	78.88 ± 4.08	75.50 ± 2.03
	UNK	/	/	74.25 ± 11.41	90.00 ± 5.08	65.50 ± 12.04	91.75 ± 2.74	43.00 ± 9.75	80.00 ± 6.06	95.50 ± 1.68	98.50 ± 1.37
	HOS	/	/	81.74 ± 7.27	90.66 ± 2.76	76.54 ± 8.97	87.06 ± 4.16	55.55 ± 8.07	72.34 ± 0.97	86.34 ± 2.40	85.46 ± 1.04
B6	OS	74.06 ± 0.22	74.19 ± 0.65	96.28 ± 1.72	95.00 ± 1.97	89.25 ± 1.80	89.88 ± 2.71	79.34 ± 7.86	78.25 ± 3.65	97.44 ± 1.80	98.56 ± 1.48
	KNO	98.75 ± 0.29	98.92 ± 0.86	97.50 ± 1.53	94.00 ± 2.54	93.17 ± 3.30	89.75 ± 3.01	79.33 ± 6.70	76.92 ± 5.53	96.58 ± 2.40	98.08 ± 1.97
	UNK	/	/	92.62 ± 3.68	98.00 ± 1.62	77.50 ± 5.93	90.25 ± 6.00	79.38 ± 14.57	82.25 ± 3.97	100.00 ± 0.00	100.00 ± 0.00
	HOS	/	/	94.97 ± 2.30	95.94 ± 1.56	84.43 ± 2.83	89.90 ± 3.35	78.98 ± 10.19	79.31 ± 2.43	98.25 ± 1.25	99.02 ± 1.02
B7	OS	65.67 ± 0.47	66.67 ± 0.00	72.25 ± 2.38	82.69 ± 7.84	81.94 ± 1.33	75.97 ± 4.06	74.28 ± 3.98	81.00 ± 1.30	98.17 ± 0.58	99.33 ± 0.30
	KNO	98.50 ± 0.71	100.00 ± 0.00	93.88 ± 2.44	91.50 ± 10.34	97.38 ± 1.35	87.12 ± 5.24	77.25 ± 3.30	84.62 ± 2.64	99.50 ± 0.28	99.50 ± 0.28
	UNK	/	/	29.00 ± 5.29	65.08 ± 14.73	51.08 ± 4.21	53.67 ± 7.18	68.33 ± 11.99	73.75 ± 1.86	95.50 ± 1.90	99.00 ± 0.63
	HOS	/	/	44.09 ± 6.18	74.93 ± 11.07	66.92 ± 3.47	66.12 ± 5.77	72.00 ± 6.30	78.77 ± 0.69	97.45 ± 0.95	99.25 ± 0.36
B8	OS	65.42 ± 1.06	66.33 ± 0.46	73.64 ± 1.71	90.08 ± 5.03	80.39 ± 1.40	74.69 ± 4.84	77.92 ± 6.01	78.81 ± 1.02	97.92 ± 0.90	99.56 ± 0.30
	KNO	98.12 ± 1.59	99.50 ± 0.68	94.00 ± 3.44	95.38 ± 3.08	95.75 ± 0.81	87.50 ± 7.28	78.88 ± 5.85	82.25 ± 1.57	99.38 ± 0.00	99.75 ± 0.34
	UNK	/	/	32.92 ± 4.37	79.50 ± 11.10	49.67 ± 4.99	49.08 ± 9.35	76.00 ± 13.09	71.92 ± 2.05	95.00 ± 2.70	99.17 ± 0.59
	HOS	/	/	48.57 ± 4.55	86.42 ± 7.59	65.28 ± 4.15	62.26 ± 7.30	76.84 ± 8.06	76.71 ± 1.11	97.12 ± 1.43	99.46 ± 0.34
Avg	OS	70.61 ± 5.16	70.52 ± 6.03	83.67 ± 9.95	90.42 ± 5.23	88.09 ± 7.22	85.24 ± 6.39	79.27 ± 6.99	82.11 ± 10.19	94.64 ± 5.55	96.16 ± 6.64
	KNO	98.04 ± 1.75	97.92 ± 4.47	95.34 ± 4.92	93.46 ± 3.72	97.48 ± 2.91	90.26 ± 4.94	84.53 ± 8.31	83.15 ± 11.01	95.86 ± 7.30	96.18 ± 8.44
	UNK	/	/	57.94 ± 26.04	84.13 ± 11.67	66.95 ± 19.68	75.59 ± 17.09	63.17 ± 18.50	80.53 ± 10.29	93.49 ± 7.93	97.50 ± 4.47
	HOS	/	/	68.55 ± 20.37	87.87 ± 7.89	77.77 ± 13.67	80.93 ± 10.87	70.23 ± 14.58	81.49 ± 10.00	94.18 ± 5.15	96.51 ± 5.04

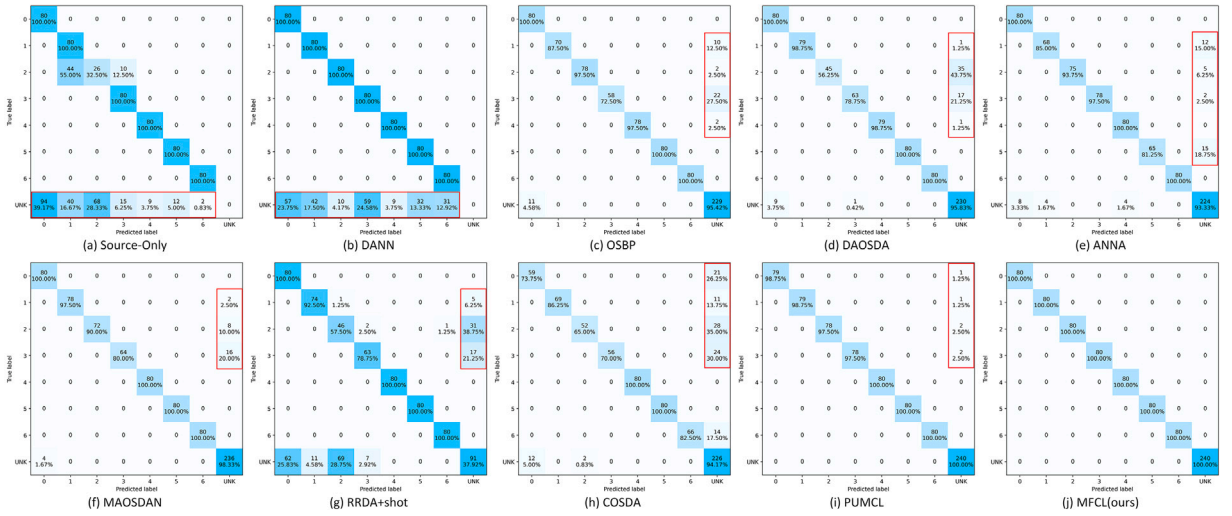


Fig. 7. Confusion matrices of various methods evaluated on the S3 task.

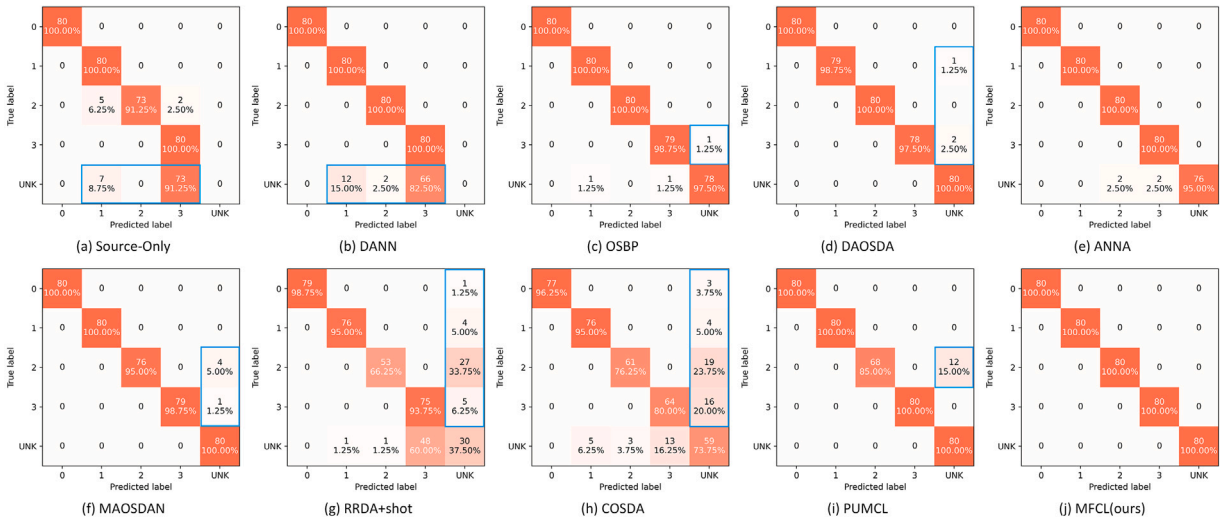


Fig. 8. Confusion matrices of various methods evaluated on the B1 task.

4.4.2. Visualization analysis

To provide deeper insight into the classification performance of each approach, confusion matrices corresponding to tasks S3 and B1 are presented in Fig. 7 and Fig. 8, respectively. In each matrix, each cell indicates the number of samples predicted as a given class (columns) versus the actual class (rows). After row-wise normalization, the diagonal cells display both the number and percentage of correctly classified samples within each class.

Based on the analysis of Figs. 7 and 8, several key observations can be made. Closed-set UDA methods, lacking a separation-driven adversarial learning module, are unable to identify unknown categories and tend to misclassify unknown samples as various known classes to different extents. In contrast, open-set UDA approaches, such as OSBP, DAOSDA, ANNA, MAOSDAN, RRDA+shot, COSDA, PUMCL, and the proposed MFCL, demonstrate better capability in detecting unknown fault types. However, it is noteworthy that while OSBP, DAOSDA, ANNA, MAOSDAN, COSDA, and PUMCL achieve relatively high accuracy in identifying unknown categories, they still misclassify a portion of known samples as unknown. This observation is consistent with the conclusions drawn in the quantitative analysis section, where the trade-off at the decision boundary between known and unknown categories can reduce recognition performance on known classes. In comparison, the proposed method, supported by self-supervised contrastive learning and class-aware consistency learning, not only significantly improves the recognition of unknown classes but also effectively avoids misclassifying known categories as unknown. This results in a more stable and balanced classification performance across categories.

To provide a visual comparison of the classification effectiveness among various open-set UDA approaches, t-SNE is utilized to project the convolutional features from the backbone network's final layer. The t-SNE visualizations for the source-only method

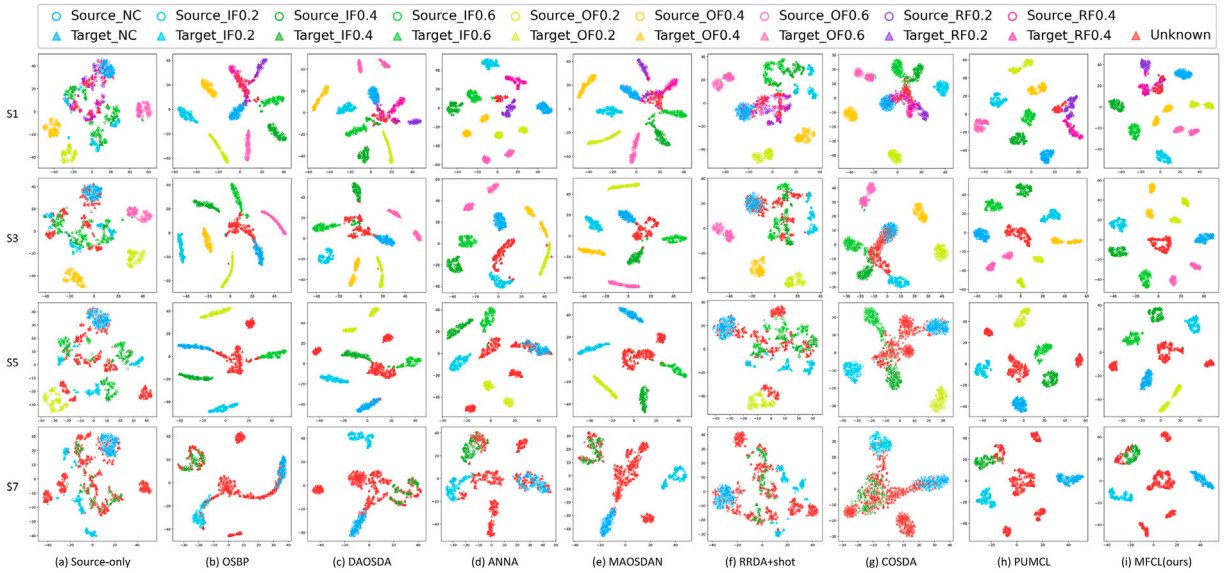


Fig. 9. t-SNE visualizations of different methods on tasks S1, S3, S5, and S7.

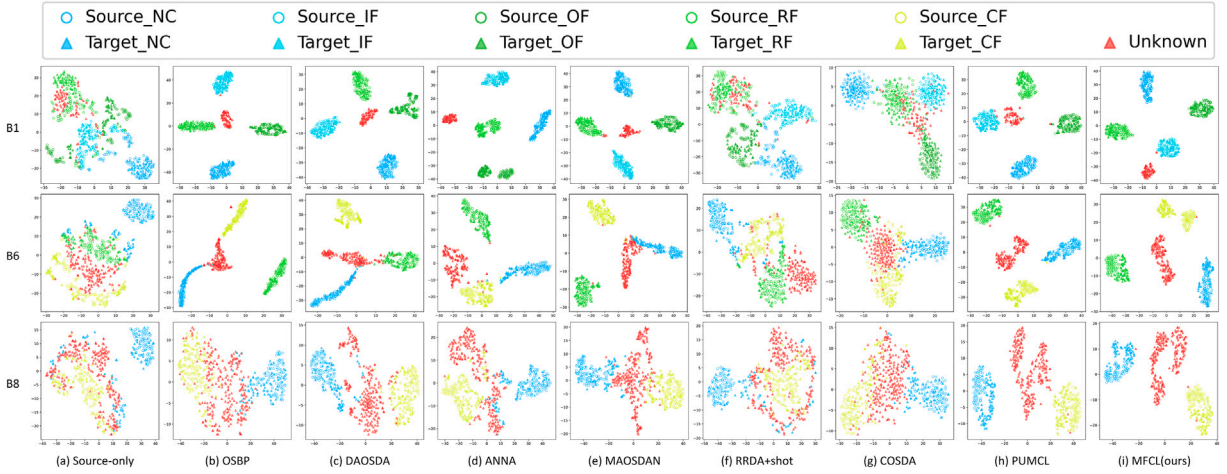
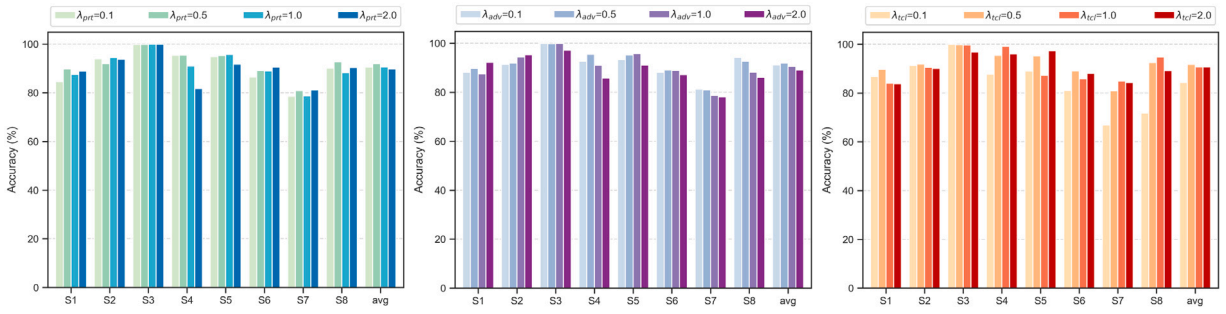


Fig. 10. t-SNE visualizations of different methods on tasks B1, B6, and B8.

alongside five representative open-set UDA approaches are illustrated in Figs. 9 and 10. Closed-set UDA methods are excluded from the comparison, as they are incapable of identifying unknown classes, resulting in poor representation of such samples. In the t-SNE plots, every point corresponds to a sample projected from the high-dimensional feature space into two dimensions. These projections effectively reflect the local relationships among samples after dimensionality reduction. This visualization enables observation of how well different adaptation methods cluster known categories and separate unknown ones. Ideally, a well-performing method should yield compact clusters for known classes while clearly distinguishing unknown categories.

It can be observed that, due to domain discrepancies, the source-only method shows suboptimal clustering performance, causing significant class overlap. Methods such as OSBP, DAOSDA, ANNA, MAOSDAN, and COSDA alleviate inter-domain distribution shifts to some extent by generating feature representations that tend to cluster. These approaches can effectively facilitate the grouping of known categories. However, they struggle to establish clear decision boundaries for all classes, resulting in considerable confusion between unknown and certain known classes and ultimately leading to misclassifications. In contrast, both PUMCL and the proposed MFCL method demonstrate superior class separability, with more compact intra-class clusters and sharper inter-class boundaries. For instance, in the S1 adaptation task shown in Fig. 9, the unknown category (RF0.6) and the known categories RF0.2 and RF0.4 all represent rolling element faults with different severity levels, leading to inherent similarities in the feature space. Consequently, RF0.2 and RF0.4 samples are highly prone to being misclassified as unknown faults. Nevertheless, compared with other methods, MFCL achieves better differentiation among these fault types, exhibiting a more pronounced class separation trend. Furthermore,



(a) Sensitive analysis of λ_{prt} (b) Sensitive analysis of λ_{adv} (c) Sensitive analysis of λ_{tcl}

Fig. 11. HOS (%) of MFCL on SDUST dataset under different hyper-parameters of λ_{prt}, λ_{adv} and λ_{tcl}.

the visualization results on the BJTU-RAO dataset, particularly in sub-tasks B6 and B8 as shown in Fig. 10, further confirm the effectiveness of the proposed method in identifying and isolating unknown categories. Overall, t-SNE visualizations provide intuitive evidence for analyzing the learned feature representations and offer qualitative assessments of classification performance in open-set UDA fault diagnosis. Nonetheless, as illustrated in the S7 task in Fig. 9, even the MFCL method shows some degree of feature overlap between known and unknown categories. Moreover, compared with closed-set models, the MFCL method still exhibits a performance gap in classifying known categories, indicating that there remains significant room for improving the effectiveness of open-set UDA approaches.

4.5. Ablation studies

4.5.1. Sensitive analysis

In this section, a systematic ablation study is conducted on the five key trade-off parameters in Eq. (16), namely λ_{sal}, λ_{ucl}, λ_{prt}, λ_{adv}, and λ_{tcl}. Considering the higher sample complexity and broader openness metric coverage of the SDUST dataset, eight representative transfer tasks from this dataset are selected for analysis. The HOS metric is adopted to evaluate the model's performance with different hyperparameters, and the associated experimental outcomes are illustrated in Figs. 11 and 12.

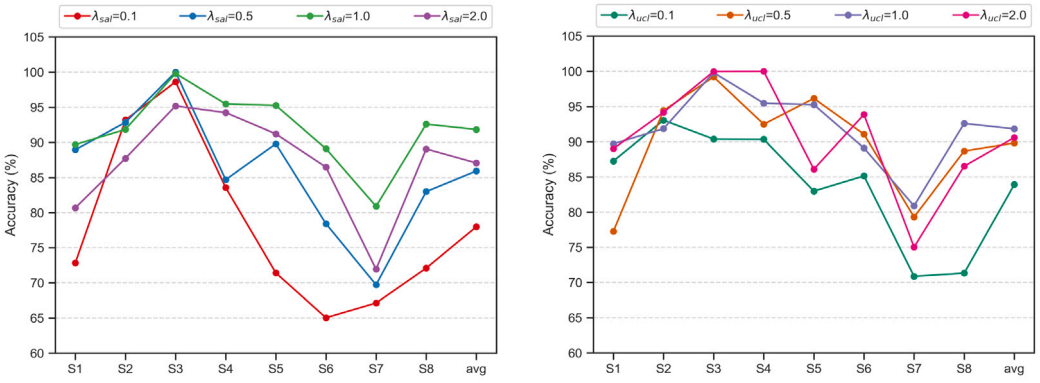
To investigate the trade-off parameters of the three losses in the FCM module, λ_{adv} and λ_{tcl} are first fixed at 0.5, and the impact of λ_{prt} is evaluated over the range from 0.1 to 2.0. As shown in Fig. 11(a), the model's performance remains relatively stable across the tested range, with a slight improvement observed at $\lambda_{prt} = 0.5$, indicating a slight advantage. Subsequently, λ_{prt} is kept at 0.5 and λ_{tcl} at 0.5, while λ_{adv} is varied from 0.1 to 2.0 to assess its impact on performance. The results in Fig. 11(b) indicate that the model achieves the best performance when λ_{prt} is set to 0.5, demonstrating the effectiveness of this configuration under the given tasks. Further analysis is conducted on the sensitivity of λ_{tcl} in the range of 0.1 to 2.0, with $\lambda_{prt} = 0.5$ and $\lambda_{adv} = 0.5$ held constant. As illustrated in Fig. 11(c), a significant performance drop is observed when $\lambda_{tcl} = 0.1$. Although both $\lambda_{tcl} = 1.0$ and $\lambda_{tcl} = 2.0$ yield strong results, setting λ_{tcl} to 0.5 achieves slightly higher HOS values, indicating its noticeable impact on model performance. Based on these findings, the configuration $\lambda_{prt} = 0.5$, $\lambda_{adv} = 0.5$, and $\lambda_{tcl} = 0.5$ is identified as a balanced and robust choice.

Similarly, for the two trade-off parameters in the OSIM module, λ_{ucl} is first fixed at 1.0, and the effect of varying λ_{sal} from 0.1 to 2.0 on model performance is analyzed. The results in Fig. 12(a) indicate that the model achieves optimal performance when λ_{sal} is set to 1.0, demonstrating the effectiveness of this configuration for the given task. Subsequently, λ_{sal} is fixed at 1.0, and the impact of λ_{ucl} is evaluated over the same range. As shown in Fig. 12(b), the performance of the model remains relatively stable across the tested range, with a slight improvement at $\lambda_{ucl} = 1.0$, suggesting a modest advantage for this setting.

This setting ensures consistent performance across multiple tasks and enhances the reliability of the experimental results by reducing the risk of bias or instability caused by suboptimal hyperparameter selection.

4.5.2. Effectiveness analysis of key modules

Building upon the optimal hyperparameter configuration, an ablation study was further carried out to evaluate the individual impact of each module on the overall performance of the model, as summarized in Table 5. The findings demonstrate that the model achieves the highest performance, with an HOS of 91.83%, when all modules are jointly utilized. Specifically, eliminating the $\mathcal{L}_{sal}$ loss results in a complete loss of the model's capacity to identify unknown categories. Excluding the $\mathcal{L}_{ucl}$, $\mathcal{L}_{prt}$, $\mathcal{L}_{adv}$, and $\mathcal{L}_{tcl}$ losses individually leads to HOS drops of 10.53%, 3.14%, 1.33%, and 11.86%, respectively. These findings clearly underscore the critical role of each component in improving the model's effectiveness. In particular, the $\mathcal{L}_{ucl}$ and $\mathcal{L}_{tcl}$ losses play a pivotal role in boosting the overall performance. The ablation experiment results show that the synergistic integration of all modules provides an effective solution for cross-domain open-set fault diagnosis tasks. Although the complete configuration yields the highest overall accuracy and harmonic mean, it does not necessarily result in the best performance for either known class classification or unknown class identification when evaluated independently. Nevertheless, this configuration maintains a favorable balance between known and unknown class performance, demonstrating strong generalization and practical applicability.



(a) Sensitive analysis of λ_{sal} (b) Sensitive analysis of λ_{ucl}

Fig. 12. HOS (%) of MFCL on SDUST dataset under different hyper-parameters of λ_{sal} and λ_{ucl}.

Table 5
Ablation study on the effectiveness of each loss component for the SDUST dataset. The best results are highlighted in bold.

Task	w/o $\mathcal{L}_{sal}$	w/o $\mathcal{L}_{ucl}$	w/o $\mathcal{L}_{prt}$	w/o $\mathcal{L}_{adv}$	w/o $\mathcal{L}_{tcl}$	MFCL
S1	/	76.33 ± 4.95	85.03 ± 4.10	86.74 ± 3.13	78.54 ± 7.14	**89.70 ± 0.78**
S2	/	78.69 ± 1.91	90.94 ± 2.10	91.22 ± 3.89	90.07 ± 1.87	**91.84 ± 2.22**
S3	/	81.48 ± 3.63	98.85 ± 2.08	99.51 ± 0.46	**99.95 ± 0.08**	99.80 ± 0.23
S4	/	84.32 ± 6.53	91.03 ± 12.28	**95.51 ± 10.04**	85.91 ± 8.36	95.47 ± 10.02
S5	/	89.35 ± 5.51	91.54 ± 13.45	91.62 ± 7.97	77.47 ± 8.88	**95.26 ± 5.91**
S6	/	80.56 ± 7.99	83.63 ± 15.38	**91.93 ± 4.52**	71.27 ± 13.18	89.10 ± 7.52
S7	/	79.53 ± 2.88	78.64 ± 7.25	79.12 ± 6.67	66.37 ± 10.36	**80.89 ± 4.84**
S8	/	80.10 ± 5.62	89.86 ± 3.23	88.36 ± 6.70	70.15 ± 4.52	**92.59 ± 5.11**
Avg	/	81.30 ± 3.98	88.69 ± 6.13	90.50 ± 6.08	79.97 ± 11.35	**91.83 ± 5.61**

Table 6
Computation time (s) of different methods.

Method		Source-only	DANN	OSBP	DAOSDA	ANNA	MAOSDAN	RRDA+shot	COSDA	PUMCL	MFCL
SDUST	Training ↓	**115.66**	281.28	254.21	767.15	492.93	669.77	121.17	718.46	1328.49	1370.16
	Inference ↓	0.10	0.10	0.09	0.09	0.11	0.09	**0.08**	0.12	0.10	0.10
BJTU-RAO	Training ↓	113.57	275.46	252.99	769.41	260.01	665.60	**57.75**	389.82	1217.88	1252.19
	Inference ↓	**0.02**	**0.02**	**0.02**	**0.02**	0.03	**0.02**	**0.02**	0.04	**0.02**	**0.02**

4.5.3. Computational efficiency analysis

Table 6 presents the average training and inference times of different methods on the SDUST and BJTU-RAO datasets. The inference time is defined as the total prediction time for all test samples. As indicated by the results in the table, the proposed method and PUMCL incur higher training costs than other approaches, owing to the computation of multiple augmented views for both the source and target domains. Nevertheless, considering the superior ability of the proposed method to detect unknown classes, the additional training overhead is well justified. In terms of inference, all methods complete prediction within one second, exhibiting no significant differences.

5. Conclusion

This work presents a robust open-set UDA framework designed for cross-domain mechanical fault diagnosis. The method effectively extracts class-specific transferable features and latent distribution structures from unlabeled target samples, thereby enhancing the model's capacity to identify unknown faults. Specifically, it integrates source prototype constraints, domain adversarial learning, and target sample multi-view contrastive learning to achieve effective alignment of feature distributions between domains and improve the representation quality of target samples from multiple perspectives. The introduction of a separation adversarial mechanism enables the model to detect and recognize unknown classes during adaptation, while the class-aware consistency learning strategy further enhances inter-class discriminability. These key components significantly enhance the model's transferability and robustness under complex working conditions with distribution shifts, marking a notable advancement in open-set UDA for mechanical fault diagnosis. To comprehensively evaluate the proposed method, comparative experiments were carried out against two closed-set UDA approaches and four state-of-the-art open-set UDA approaches. Experimental outcomes on the SDUST

and BJTU-RAO datasets highlight that the proposed method achieves the highest OA and HOS scores in 10 out of the 16 transfer tasks, significantly outperforming existing open-set UDA approaches.

Further evidence from confusion matrices and t-SNE visualizations confirms the method's superiority in improving class boundary clarity and unknown class clustering quality, especially when unknown faults exhibit low overlap with known categories. Additionally, ablation studies reveal significant performance drops when key modules are removed, validating the effectiveness of the designed loss functions in boosting diagnostic accuracy.

CRediT authorship contribution statement

Guiping Chen: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation. **Dong Xiang:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Tingting Liu:** Writing – review & editing, Validation, Conceptualization. **Feng Xu:** Validation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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